

# PROGRAMMING & SYSTEM DEVELOPMENT (M)

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# Aims

- To enhance existing skills and practical processes for programming in Python and Java, which are the main languages used in the MSc programme
- To improve skill at using online information to pick up new languages, APIs or tools
- To become familiar with common tools needed to build working systems, including repositories, virtual machines and servers.
- To develop knowledge of the use of the LINUX operating system and experience of using the tools available under LINUX for programming system administration, and cybersecurity.

# Learning Outcomes

1. Demonstrate familiarity with the relevant modern programming environments;
2. Use the command line and a text editor competently, ideally with experience with a modern IDE;
3. Develop well-structured programs in Java and Python, and be proficient in looking for online help, and be prepared to pick up new skills independently;
4. Explain the importance of concise and informative documentation, and use modern documentation tools;
5. Provide evidence of program correctness through thorough testing;

# Learning Outcomes

6. Describe and use version control processes, and be able to checkout and use a repository (e.g. git);
7. Launch, edit and use Jupyter notebook, Colab, Conda.;
8. Use basic SQL from Java and Python, and be able to set up a basic server;
9. Describe the basics of user interaction with the Linux shell, file navigation, system monitoring, printing;
10. Explain how to install and configure software under Linux, and be able to write and execute simple Linux shell scripts..

# Lectures' contents

1. Introduction
2. Python Basics, Decision Making & Object-Oriented Programming (OOP) in Python, Intro to Conda, Google Colab
3. Python – Repetition, functions, Data structures
4. Python – Files & Exception, SQLite3, PyTorch
5. Data Processing – Pandas , Numpy
6. Data Visualisation - Matplotlib
7. Java and Unix
8. Java and Unix
9. Java and Unix
10. Java and Unix
11. Java and Unix
12. Revision week

# Assessments

- Tutorial exercises
  - Sample solutions available.

The graded assessments and their weighting in this course are as follows:

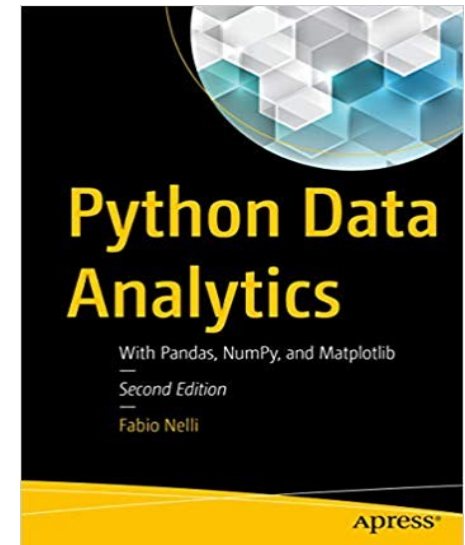
- **Participation in peer review (tAPP):** weighting 0.05 (5%)
- **Assessed Exercise (invigilated):** Broken down into the following exercises
  1. Assessed Exercise 1 (**10%** - Python)      Date: 03/10/2025
  2. Assessed Exercise 2 (**15%** - Python)      Date: 10/10/2025
  3. Assessed Exercise 3 (**15%** - Java **OR** Unix)      Date: 24/10/2025

Total weight for Assessed Exercises: weighting **0.40 (40%)**

- **Programming Assessment:** weighting **0.15 (15%)**      Date: 31/10/2025
- **Lab Exam – Date: TBC – On campus and invigilated**
  - weighting **0.4.** (40%)

# Reading (1)

- Fabio Nelli  
***Python Data Analytics***  
Apress (2018)
- Jhon Dierre  
***Linux Shell - Scripting with Bash***
- Paul Smart  
***Shell Scripting and Advanced Linux Commands***  
***Beginners Guide***



# Reading (2)

- <http://www.dcs.gla.ac.uk/bootcamp/>
  - Username “student”
  - Password “Letmein”



# Modified Team-Based Learning Approach

- **Pre-Lecture material**
  - Read the material before coming to the lecture
  - Take the quiz or any test
- **Lecture Material – Problem Solving Tasks (1 hour per session)**
  - You are given some problems to solve based on what you have read
  - Try to solve the problems alone first
  - Then discuss with your team members.
  - Produce a solution as a team
  - Give your solution to another team for peer – evaluation
- **Tutorial/Labs (2 hour per session)**
  - Support each other.

# Other Things to know

- **Tutorials/Labs:** released at least a week before to give you the chance to attempt before the next class. Solutions some time after!
- Always have the following items with you in lectures:
  - Paper
  - Pen/pencil

# Important note about lab groups

- Lab sections 1, 2, 3 and 4 are divided into teams.
- Each student is a member of one team.
- Students within the same team:
  - May support each other during the labs
  - Participate in the group peer review of team Application exercises (tAPP)

Any Question?